Fatemeh Shahnazian

Power Systems Engineer | PhD | Hybrid Power Plants & Weak-Grid Stability

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PROFILE

Power systems engineer with a PhD from DTU and 10 years of combined industry and research experience in grid integration of converter-based generation. Specialised in EMT/RMS modelling (PSCAD, MATLAB, PowerFactory), control design for hybrid power plants, and grid-code compliance in weak-grid conditions. Author of 11 peer-reviewed papers, with hands-on experience validating OEM equipment models and working with international partners and grid operators. Based in Denmark and seeking a long-term role in wind, hybrid power plant, or grid-integration engineering.

TECHNICAL SKILLS

Power systems	Hybrid power plants, weak-grid stability, voltage and frequency control, grid-forming vs. grid-following control, MMC-based systems, low-inertia systems
Studies & analysis	EMT and RMS dynamic studies, load flow, fault ride-through (FRT), power quality, eigenvalue and root-locus analysis, weak-grid risk assessment
Grid codes & standards	ENTSO-E network codes, Energinet grid connection requirements, IEC 61000, IEC 61400
Tools	PSCAD/EMTDC, MATLAB/Simulink, DIgSILENT PowerFactory; basic Python

PROFESSIONAL EXPERIENCE

Researcher in Wind and Energy Systems (PhD) — **2021 – 2025**

DTU Wind and Energy Systems, Roskilde, Denmark

- Developed and validated detailed PSCAD EMT models of hybrid renewable power plants, including converter-based resources, plant controllers and grid interfaces, to assess stability and control performance in weak grids.
- Ran system-level dynamic studies (PSCAD, MATLAB; RMS/EMT) to identify critical stability challenges and propose mitigation strategies for large-scale wind and offshore renewable integration.
- Assessed grid-code compliance of converter-based plants under faults and disturbances across multiple operating scenarios.
- Parameterised, benchmarked and consistency-checked converter and plant controller models under fault and weak-grid conditions; approach directly transferable to OEM and vendor-supplied models.
- Contributed to IEA Wind Task activities and completed a research stay at the University of Queensland, working with industry experts and grid operators so that modelling assumptions reflected practical connection requirements.

Electrical Engineer, Power / Maintenance — **2015 – 2021**

Raymand Petro Energy Company (RPEC), Tehran, Iran

- Performed power system analysis and troubleshooting for industrial generation systems in onshore and offshore facilities.
- Built and validated RMS and EMT models from manufacturer generator and transformer data; verified OEM model response under fault scenarios.
- Simulated generator operation, stability and control in grid-connected and islanded conditions; analysed fault and switching transients, power quality and system reliability.
- Supported maintenance and overhaul of generators, drives and transformers.

Power System Researcher — **2017 – 2021**

Babol Noshirvani University of Technology, Babol, Iran

- Developed control algorithms and dynamic models for converter-based systems to improve frequency stability in low-inertia conditions.
- Designed automatic controller-tuning methods based on target dynamic performance, supported by eigenvalue and root-locus analysis.
- Developed models and controls for MMC-based systems, reducing circulating currents and improving stability and efficiency; analysed harmonics and transient behaviour of power electronic systems.
- Tracked project timelines and deliverables and coordinated with external project partners.

SELECTED PUBLICATIONS

11 peer-reviewed papers and a PhD thesis; 85+ citations. Google Scholar profile

- **Hybrid power plant control to enhance voltage stability in weak grid.** *Int. Journal of Electrical Power & Energy Systems 172, 2025.*

- **Aspects of relevance of hybrid power plants in control and stability of weak grids.** *Energies 17(24), 2024.*
- **Hybrid power plant control in weak grids: grid following vs. grid forming.** *23rd Wind & Solar Integration Workshop (WIW), 2024.*
- **Challenges of renewable energy integration to weak grids.** *IET Conference Proceedings CP847, 2023.*
- **Interfacing modular multilevel converters for grid integration of renewable energy sources.** *Electric Power Systems Research 160, 2018.*
- **Control Systems for Hybrid Power Plants in Weak Grids (PhD thesis).** *DTU Wind and Energy Systems, 2025.*

EDUCATION

PhD, Electrical Engineering – Power Systems **2021 – 2025**
Technical University of Denmark (DTU), Roskilde, Denmark

MSc, Electrical Engineering – Power Systems **2014 – 2017**
Babol Noshirvani University of Technology, Iran

BSc, Electrical Engineering **2010 – 2014**
K. N. Toosi University of Technology, Tehran, Iran

LANGUAGES & INTERESTS

Languages: English – full professional proficiency; Danish – intermediate (DU3, Module 3, ongoing); Persian – native.
Interests: Travel and cultural exploration, learning Danish language and culture, crochet.